

RMIC Introduction

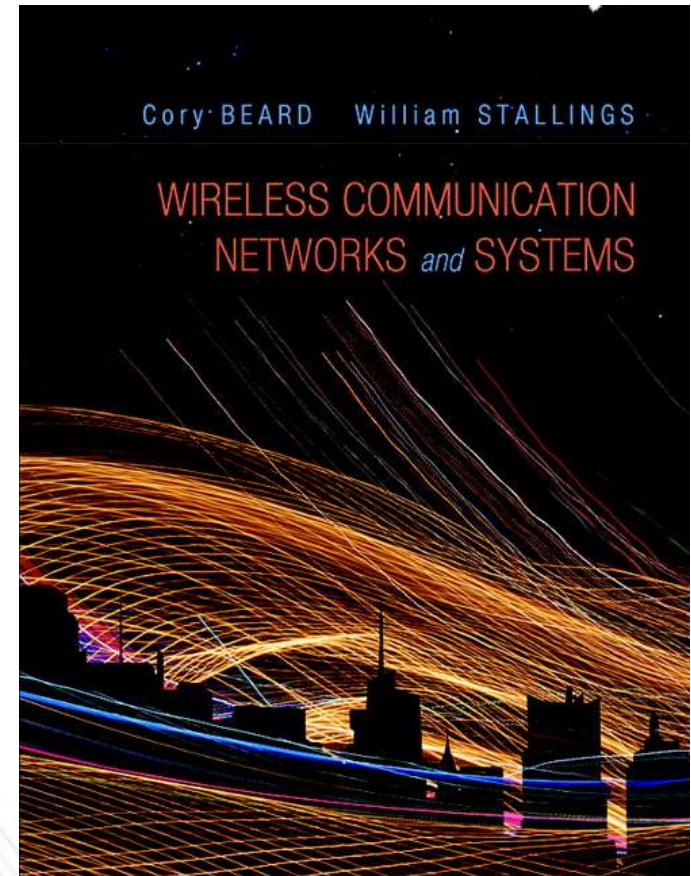
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RMIC INTRODUCTION

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Wireless Communication Networks and Systems

1st edition

Cory Beard, William Stallings

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WIRELESS COMES OF AGE

- Guglielmo Marconi invented the wireless telegraph in 1896
 - Communication by encoding alphanumeric characters in analog signal
 - Sent telegraphic signals across the Atlantic Ocean
- Communications satellites launched in 1960s
- Advances in wireless technology
 - Radio, television, mobile telephone, mobile data, communication satellites
- More recently
 - Wireless networking, cellular technology, mobile apps, Internet of Things, Internet of Everything

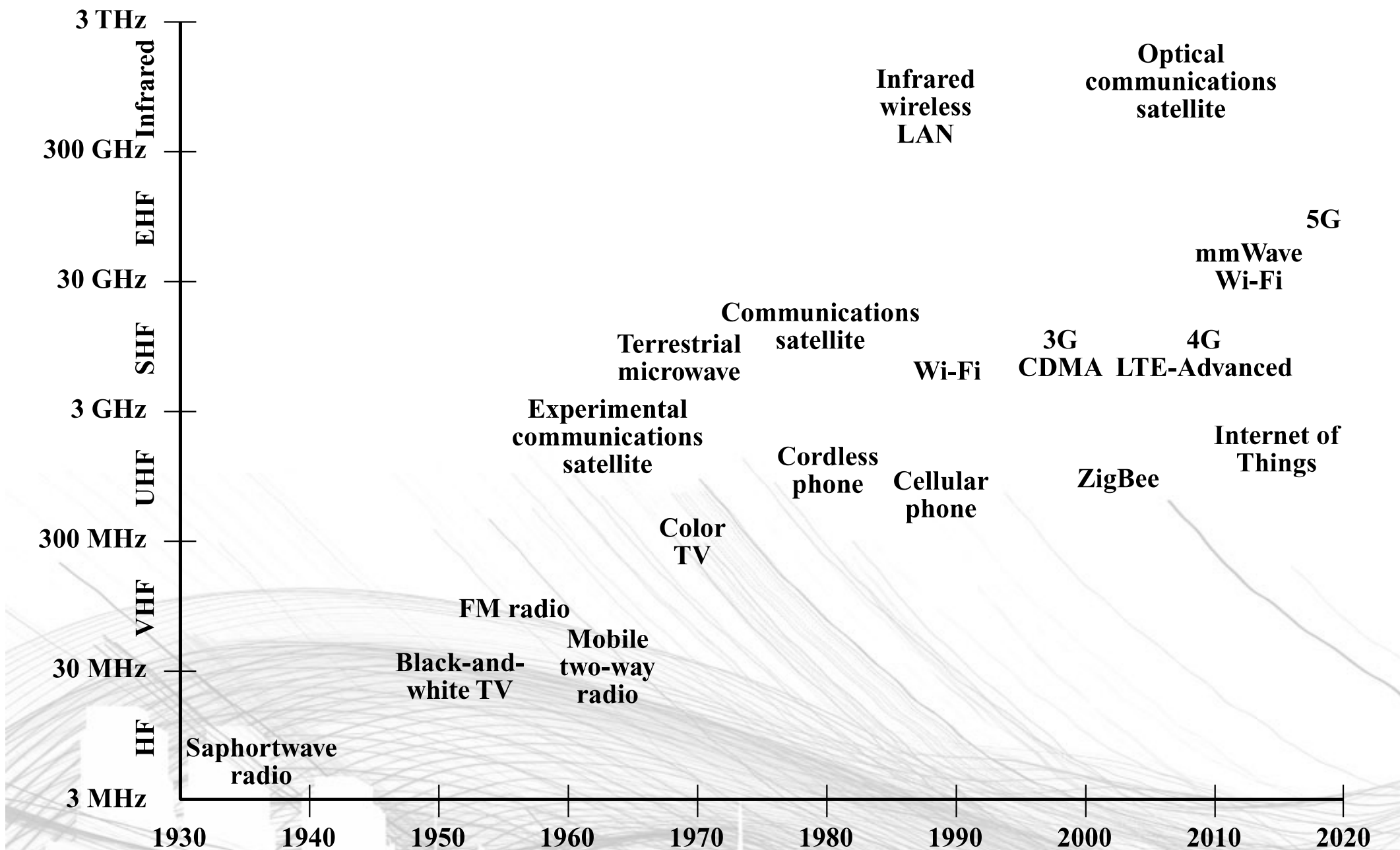


FIGURE 1.1 SOME MILESTONES IN WIRELESS COMMUNICATIONS

MOBILE DEVICE REVOLUTION

- Originally just mobile phones
- Today's devices
 - Multi-megabit Internet access
 - Mobile apps
 - High megapixel digital cameras
 - Access to multiple types of wireless networks
 - Wi-Fi, Bluetooth, 3G, and 4G
 - Several on-board sensors
- Key to how many people interact with the world around them

MOBILE DEVICE REVOLUTION

- Digital, better use of spectrum
- Decreased costs
- Limited displays and input capabilities
- Tablets provide balance between smartphones and PCs
- Long distance
 - Cellular: 4G, 5G+
 - LPWAN: LoRaWAN, SigFox, NB-IoT
- Local areas
 - Wi-Fi: Wi-Fi 6, Wi-Fi 7...
- Short distance
 - Bluetooth, ZigBee, Thread

THE TROUBLE WITH WIRELESS

- Wireless is convenient and less expensive, but not perfect
- Limitations and political and technical difficulties inhibit wireless technologies
- Wireless channel
 - Line-of-sight is best but not required
 - Signals can still be received
 - Transmission through objects
 - Reflections off of objects
 - Scattering of signals
 - Diffraction around edges of objects

THE TROUBLE WITH WIRELESS

- Wireless channel
 - Reflections can cause multiple copies of the signal to arrive
 - At different times and attenuations
 - Creates the problem of *multipath fading*
 - Signals add together to degrade the final signal
 - Noise
 - Interference from other users
 - Doppler spread caused by movement

COMBATING PROBLEMS

- Modulation – use a signal format to send as many bits as possible
- Error control coding – add extra bits so errors are detected/corrected.
- Adaptive modulation and coding – dynamically adjust modulation and coding to current channel conditions.
- Equalization – counteract the multipath effects of the channel.
- Multiple-input multiple-output systems – use multiple antennas
 - Point signals strongly in certain directions
 - Send parallel streams of data.
- Direct sequence spread spectrum – expand the signal bandwidth
- Orthogonal frequency division multiplexing – break a signal into many lower rate bit streams
 - Each is less susceptible to multipath problems.

POLITICAL DIFFICULTIES

- Between companies
 - Need common standards so products interoperate
 - Some areas have well agreed-upon standards
 - Wi-Fi, 4G, 5G
 - Not true for Internet of Things technologies
- Spectrum regulations
 - Governments dictate how spectrum is used
 - Many different types of uses and users
 - Some frequencies have somewhat restrictive bandwidths and power levels
 - Others have much more bandwidth available

CURRENT TRENDS

- 5G and gigabit Wi-Fi now being deployed, 6G under preparation
- Machine-to-machine communications
 - The “Internet of Things” (or “Internet for Things”?)
 - Devices interact with each other
 - Healthcare, disaster recovery, energy savings, security and surveillance, environmental awareness, education, manufacturing, and many others
 - Cloud Computing
 - Automated adaptation and control
 - Home sensors collaborate with home appliances, HVAC systems, lighting systems, electric vehicle charging stations, and utility companies.
 - Traffic accident prevention (autonomous cars...)
 - Eventually could interact in their own forms of social networking
 - Autonomous vehicles, robots
- SDN, Network Virtualization, Network Slicing, etc.

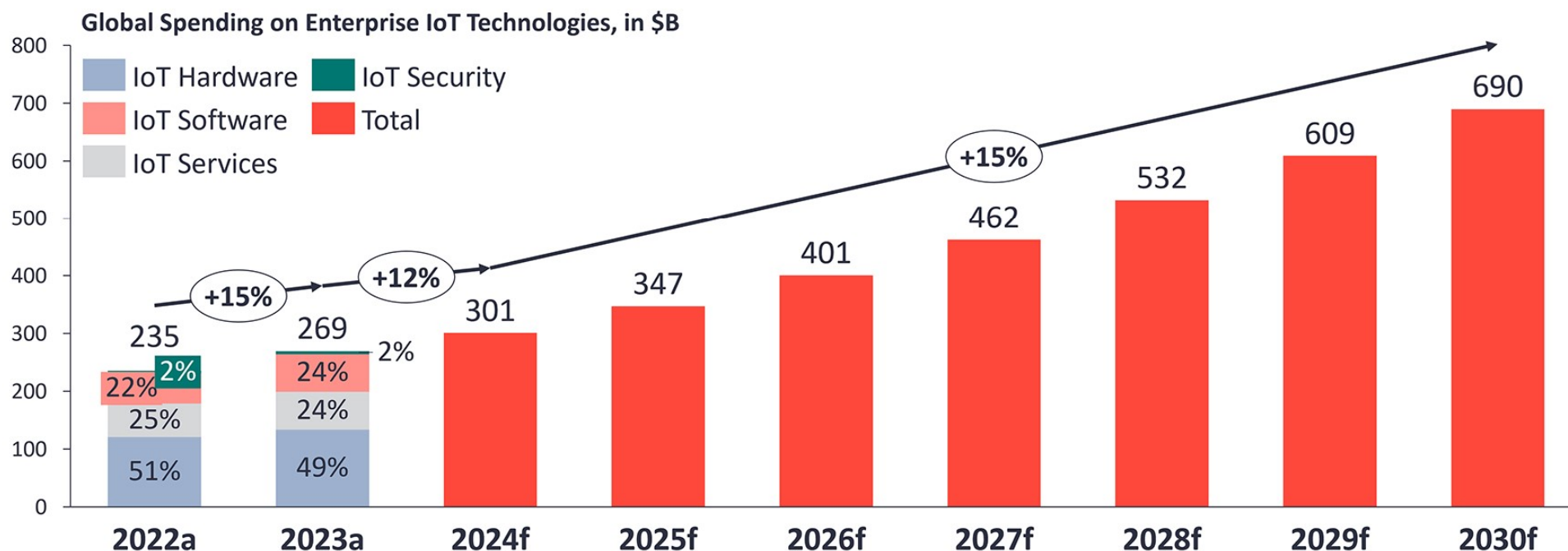
CURRENT TRENDS

- Massive Machine-to-machine communications
 - 100-fold increase in the number of devices
 - Type of communication would involve many short messages
 - Some critical control applications will have real-time delay requirements, and/or broadband requirements
 - Much more stringent than for human interaction
 - 5G+, Edge Computing

CURRENT TRENDS

The enterprise IoT market by technology 2023–2030

Data as of June 2024

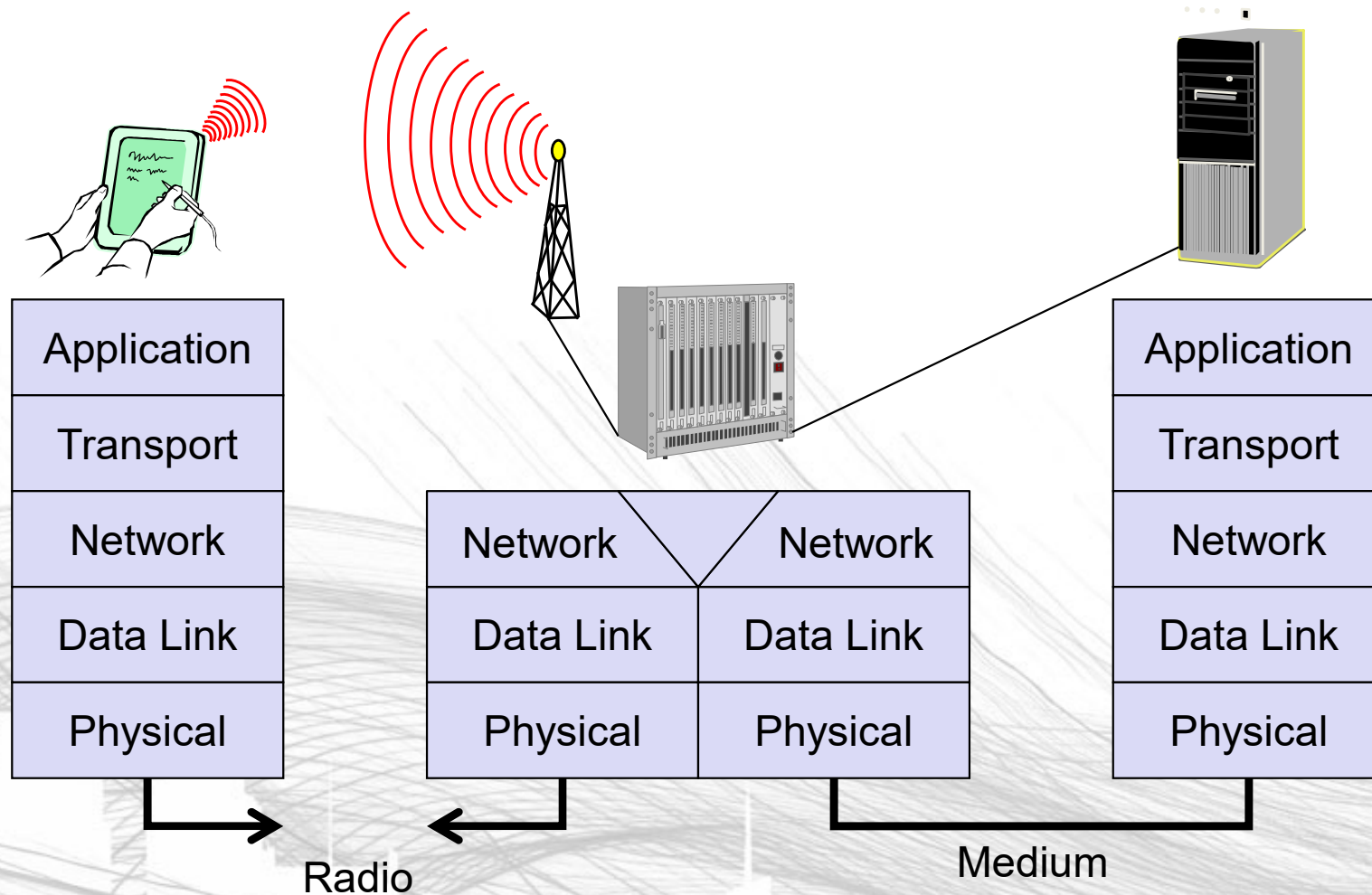


Note: IoT Analytics defines IoT as a network of internet-enabled physical objects. Objects that become internet-enabled (IoT devices) typically interact via embedded systems, some form of network communication, or a combination of edge and cloud computing. The data from IoT-connected devices is often used to create novel end-user applications. Connected personal computers, tablets, and smartphones are not considered IoT, although these may be part of the solution setup. Devices connected via extremely simple connectivity methods, such as radio frequency identification or quick response codes, are not considered IoT devices. Since the last update in 2023 our definition of the enterprise IoT tech stack slightly changed.

a: Actuals, f: Forecast

Source: IoT Analytics Research 2024 – Global IoT Enterprise Spending Dashboard (Q2/2024 update). We welcome republishing of images but ask for source citation with a link to the original post or company website.

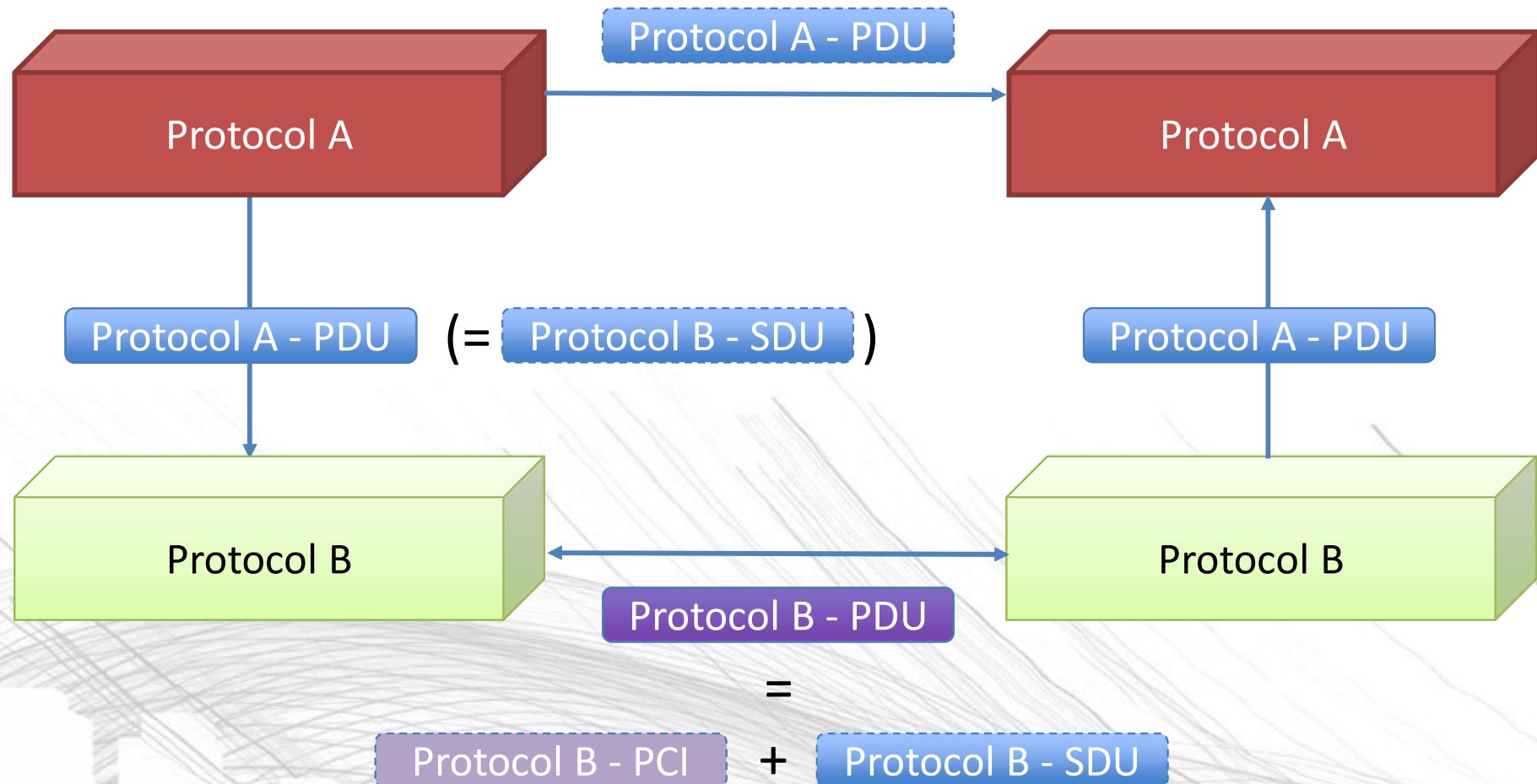
PROTOCOL REFERENCE MODEL



PROTOCOL LAYER MODEL

Application layer	service location new/adaptive applications multimedia
Transport layer	congestion/flow control quality of service
Network layer	addressing, routing device location hand-over
Data link layer	authentication media access/control multiplexing encryption
Physical layer	modulation interference attenuation frequency

PROTOCOL NOMENCLATURE



PDU = Protocol Data Unit **SDU = Service Data Unit** **PCI = Protocol Control Information**
ICI = Interface Control Information **IDU = Interface Data Unit**

RMIC - OBJECTIVES

- To convey the students the theoretical knowledge on the technological challenges of wireless and mobile communications, as well as existing ways to address them;
- To provide the students an understanding of how several basic techniques can be integrated in a wireless and/or mobile communication technology, and how the advantages and disadvantages of the latter may be analyzed in terms of functional and performance requirements;
- To provide the students an understanding of the role of communication technologies in a wider application environment of Internet of Things;
- To provide the students an experimental learning of wireless networks, by manipulating environmental and technological parameters in order to observe their impact;
- To convey the students the capability to project and build real systems to be integrated in the Internet of Things.

The first three objectives are addressed by the theoretical component, while the last two are addressed by lab and project components.

RMIC - TOC

- Introduction to RMIC
- Internet of Things
 - Concept
 - Architecture
 - High-layer protocols: CoAP, MQTT
- Principles of Wireless Communications
 - Transmission Fundamentals
 - The Wireless Channel
 - Digital Signal Modulation
 - Spread Spectrum and OFDM
 - CDMA, OFDMA
- IoT Communication Technologies
 - IEEE 802.11 (WiFi)
 - IEEE 802.15.1 (Bluetooth)
 - IEEE 802.15.4 (ZigBee, Thread)
 - LPWAN: LoRaWAN
- Cellular Technologies
 - LTE, 5G
 - (NB-IoT)